

Junhyeok Park

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Research Interests

- Mechanistic Interpretability of Large Language Models
- In-context Learning, Truthfulness, and Refusal Mechanisms
- Causal Intervention and Representation Analysis

Education

Pusan National University, Busan, South Korea

Mar 2020 – Feb 2026

B.S. in Computer Science — B.A. in French Language and Literature

GPA: 4.32/4.5 — Summa Cum Laude

Research Experience

Undergraduate Researcher, Data Intelligence Lab, Pusan National University

Sep 2025 – Feb 2026

Prof. Junsu Cho

- Contributed to experimental design and execution for emotional support conversation (ESC).
- Second author on a paper under review at EMNLP.

Research Intern, SAILAB, KAIST

Jun 2025 – Aug 2025

Prof. Jaesik Choi — Mentored by Cheongwon Kang — [report](#), [code](#)

- Analyzed response inconsistency in LLMs under prompt-format perturbations in multiple-choice QA.
- Linked output inconsistency to internal dynamics using the logit lens.

Selected Independent Research

Mechanistic Interpretability

Mar 2026 – Present

- **Zero-shot inference failure modes** [[blog](#), [code](#)]
 - Designed and conducted experiments in a zero-shot inference setting.
 - Verified the instability of the zero-shot inference setting due to prompt biases.
- **Geometry of Truth reproduction** [[blog](#), [code](#)]
 - Reimplemented PCA and causal probing (NIE) pipeline from the paper (Marks & Tegmark, 2023) without reference to existing code.
 - Verified scale-dependent emergence of linear structure and observed weak intervention effects in small models.
- **Refusal direction reproduction** [[blog](#), [code](#)]
 - Reimplemented refusal direction extraction and activation intervention pipeline from the paper (Arditi et al., 2024) without reference to existing code.
 - Verified directional control over refusal behavior.

Awards & Honors

Academic Excellence Scholarship, Pusan National University

2020 – 2025

Gochon Scholarship, Gochon Foundation

Aug 2025

Outstanding Effort & Attitude Award, Toronto Metropolitan University

Jul 2024

Languages

English (TOEFL iBT 98)

French (DELFB2)